Christian Corona

ROBOTICS | REINFORCEMENT LEARNING | SIM TO REAL

(910) 506-9190 coronachris2022@gmail.com christiancorona@ufl.edu
linkedin.com/in/christian-corona-380323390 github.com/chrisrcorona5

SUMMARY

Simulation to deployment is where I like to live. I train RL policies in sim, get them running on real robots, and work on the gap that opens up when reality doesn't match the model. Quadrupeds so far, now exoskeletons and rehab devices.

EXPERIENCE

Graduate Research Assistant, MS Thesis | University of Florida (NSF HAIC)

Gainesville, FL | Fall 2026 to Present | Advisor: Dr. Jie Fu

- Building safe RL controllers for exoskeletons and rehab devices, including a perturbation treadmill, working alongside researchers on control barrier function safety.
- Developing adaptable system identification that the RL policy depends on, aiming to beat standard domain randomization when transferring from sim to real.
- Setting up human musculoskeletal simulation with MyoSuite models in mjlab and MuJoCo, and exploring imitation learning to copy how people react to pushes and robotic assistance.

AI Engineer Intern, AI Controllers Team | IHMC Robotics Lab

Pensacola, FL | Summer 2026 | Advisor: Dr. Geoffrey Clark

- Built a LiDAR perceptive locomotion pipeline using teacher student distillation. An encoder decoder turns 3D point clouds into a short term terrain memory by reconstructing the ground truth height map, so the robot keeps working when sensing gets noisy.
- Trained one policy in Isaac Lab for the Unitree Go2 that both runs and jumps with no mode switching. Jumping is a step signal layered on the running command, with separate rewards for takeoff, flight and landing (based on arXiv:2602.00678).
- Tuned reward weights and curriculum stages so the policy stays stable across rough terrain and tracks velocity commands.
- Wrote the C++ deployment code that runs ONNX RL policies on NVIDIA Jetson, streaming motor commands and robot state over high rate Ethernet.
- Deployed on the Go2 hardware and studied the architecture of IHMC's in house biped.

PROJECT

Physics Informed Neural Network Framework | Undergraduate Capstone

Belhaven University | Advisor: Dr. Jarrod Mau | GitHub

- Built a JAX framework that bakes conservation laws into training, solving a 6 compartment SEIRD disease model from sparse data.

SKILLS

Robotics + Sim
Isaac Lab, MuJoCo, mjlab, MyoSuite, ROS2

Learning + Control
Deep RL, safe RL, system ID, policy distillation, imitation learning, sequence models, PyTorch, JAX, ONNX

Perception
LiDAR terrain mapping, perceptive locomotion

Code + Hardware
C++, C, Python, C#, NVIDIA Jetson

EDUCATION

M.S. Electrical and Computer Engineering
University of Florida
Aug 2026 to May 2028
Fall 2026: Machine Learning, Digital Signal Processing

B.S. Computer Science, Magna Cum Laude
Belhaven University
April 2026
Deep Learning, Differential Equations, Graph Theory, Number Theory

BEYOND THE LAB

Four years of college baseball as a pitcher at Freed-Hardeman and Belhaven, plus a summer in the Appalachian League (MLB sponsored). It taught me to perform under pressure, work in a team, and manage time while keeping honors grades.